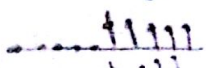


* $\delta(n) \rightarrow$ unit Sample "1 on $n=0$ only" 

* $u(n) \rightarrow$ unit Step "1 on $n \geq 0$ " 

* $u_r(n) \rightarrow$ unit Ramp "n on $n \geq 0$ " 

Section 5 - Sheet 3

1. Sketch the output sequence in each case for the following input sequence x

$\{ \dots, 0, 1, 1, 1, 1, \frac{1}{2}, \frac{1}{2}, 0, 0, \dots \}$

$\uparrow$

a) $x(n-2)$

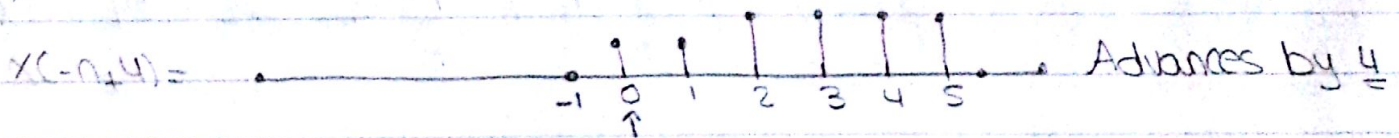
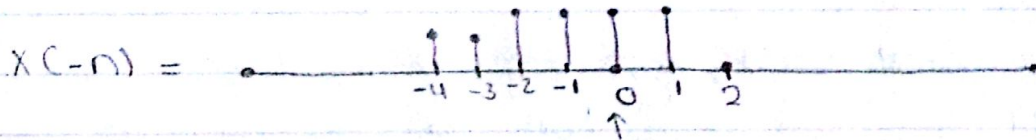
$= \{ \dots, x(-3), x(-2), x(-1), x(0), x(1), x(2), x(3), x(4), \dots \}$

$\uparrow$

$= \{ \dots, 0, 0, 1, 1, 1, 1, \frac{1}{2}, \frac{1}{2}, \dots \}$

$\uparrow$

b) $x(-n+4)$



c) $x(n+2)$

$= \{ \dots, x(-2), x(-1), x(0), x(1), x(2), x(3), x(4), x(5), \dots \}$

$\uparrow$

$= \{ \dots, 0, 1, 1, 1, \frac{1}{2}, \frac{1}{2}, 0, \dots \}$

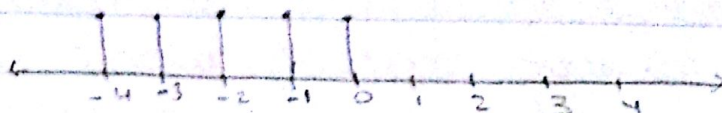
$\uparrow$

$$x(n) = \{ \dots, 0, \frac{1}{2}, 1, 1, 1, \frac{1}{2}, \frac{1}{2}, 0, \dots \}$$

d) $x(n) \cdot u(9-n)$

$$u = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

$$\rightarrow u(-n) =$$



$$\rightarrow u(9-n) =$$

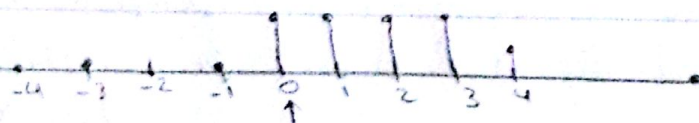


$$\rightarrow x(n) \cdot u(9-n) = \{ \dots, 0, 1, \frac{1}{2}, 1, 1, 0, 0, \dots \}$$

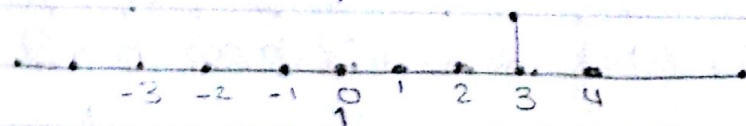
e) $x(n-1) \cdot \delta(n-3)$

$$\delta = \begin{cases} 1 & n=0 \\ 0 & n \neq 0 \end{cases}$$

$$x(n-1) =$$



$$\delta(n-3) =$$



$$x(n-1) \cdot \delta(n-3) = \{ \dots, 0, 0, 0, 0, 1, 0, \dots \}$$

f) $x(n^2)$

$$= \{ \dots, x(4), x(1), x(0), x(1), x(4), \dots \}$$

$$\quad \quad \quad -2 \quad -1 \quad 0 \quad 1$$

$$= \{ \dots, \frac{1}{2}, 1, 1, 1, \frac{1}{2}, \dots \}$$

